AI-Powered Talent sourcing & candidate ranking system

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Agenda

- Executive Summary
- Business Problem
- Exploratory Data Analysis
- Text pre-processing techniques
- Text vectorisation methods used & results
- Deployment/ Demo
- Tools used & Skills demonstrated
- Future improvements
- Conclusion

Executive Summary

We developed a system that can identify and rank candidates effectively. Rankings can be improved using human feedback

- **Lexical Models: Bag of Words (BoW) and TF-IDF (Term Frequency-Inverse Document Frequency)**
- **Word Embedding Models: Word2Vec, GloVe, and FastText**
- **Contextual Embedding Models: BERT and SBERT**

Each method is used to rank candidates based on cosine similarity to a search query, and their performance is comparatively analyzed.

- **SBERT and BERT (Generic): These models consistently demonstrated superior performance in capturing semantic relevance(context).**

Client: Talent sourcing & Mgt Co

Problem

- Finding talented candidates is not easy:
    - Understand what the client wants in a potential candidate
    - What makes a candidate shine in the role we are searching for?
    - Where to find the potential candidate?

HR challenge

How can we automatically:

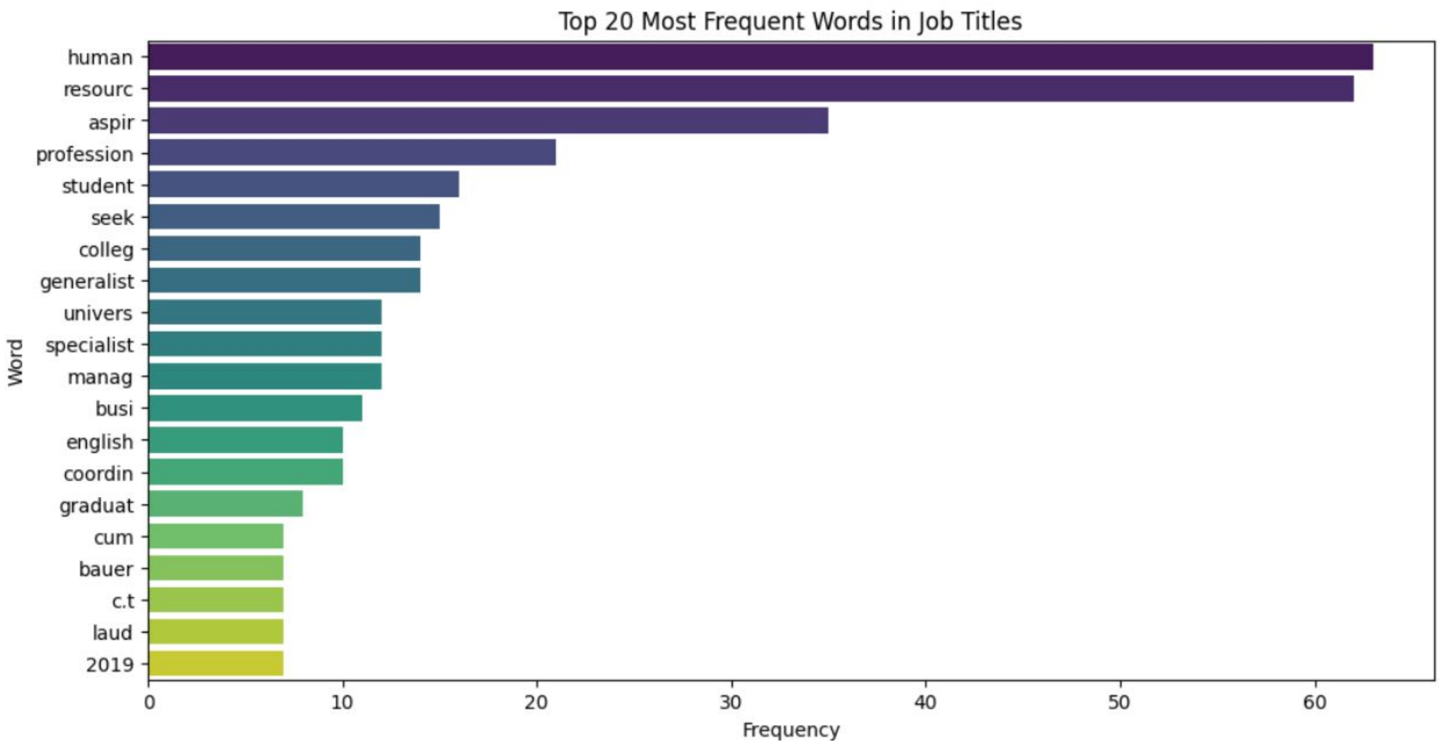
- identify the best candidates for a role
- Rank them effectively
- continuously improve rankings using human feedback

Dataset: 104 records

Attributes:

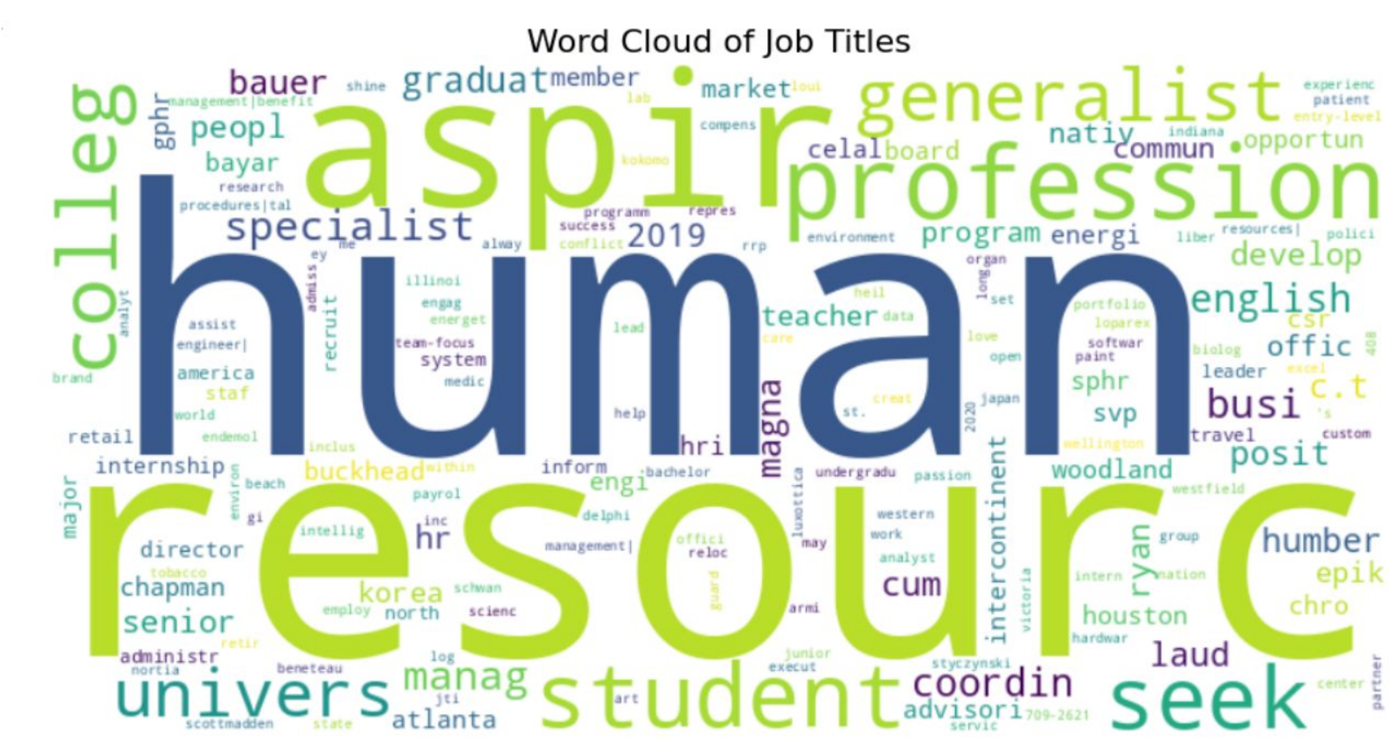
- id : unique identifier for candidate (numeric)
- job_title : job title for candidate (text)
- location : geographical location for candidate (text)
- connections: number of connections candidate has, 500+ means over 500 (text)
- fit - how fit the candidate is for the role? (numeric, probability between 0-1)

EDA: Most Frequent Words in Job Titles



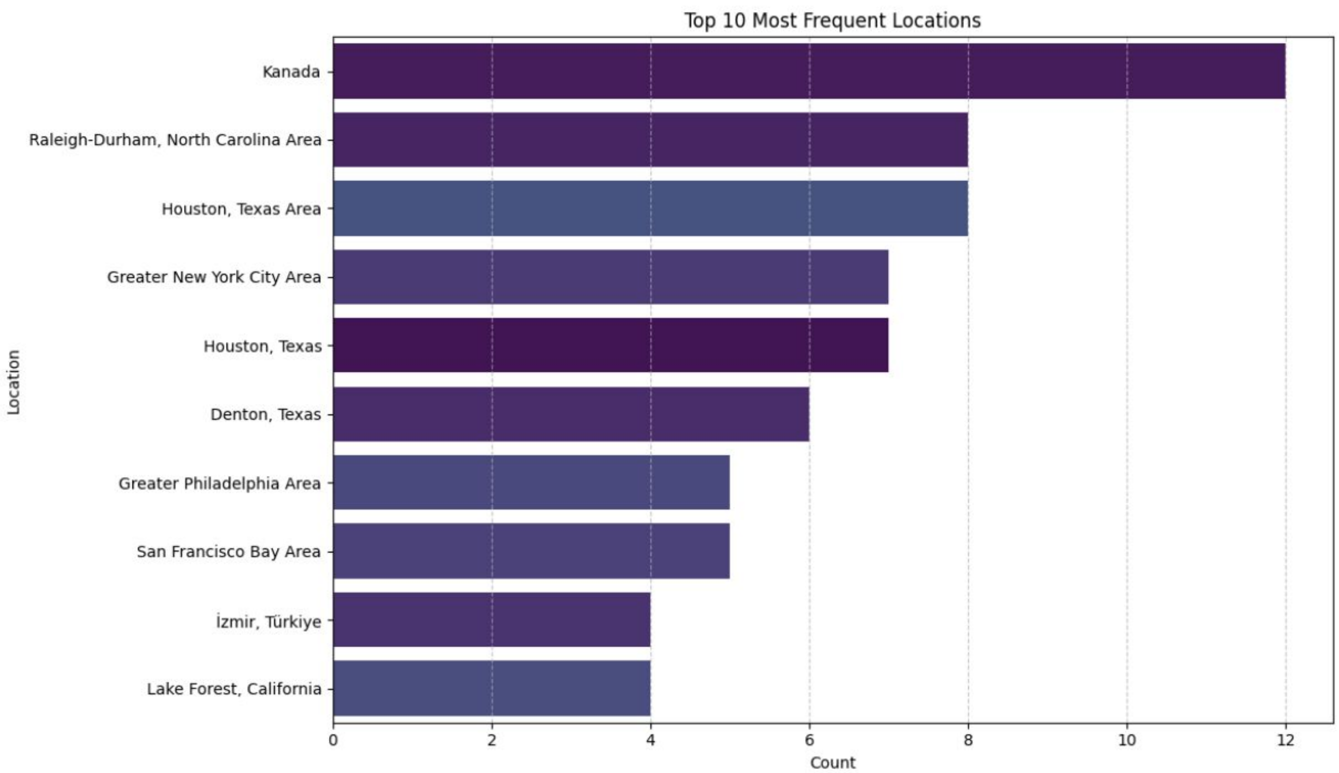
- **Dominance of 'Human Resources' related terms**
- **Common Modifiers and Aspirations:** aspir, profession, student
- **Educational and Business Context:** Terms like 'colleg', 'univers', and 'busi'

EDA: Most Frequent Job Title



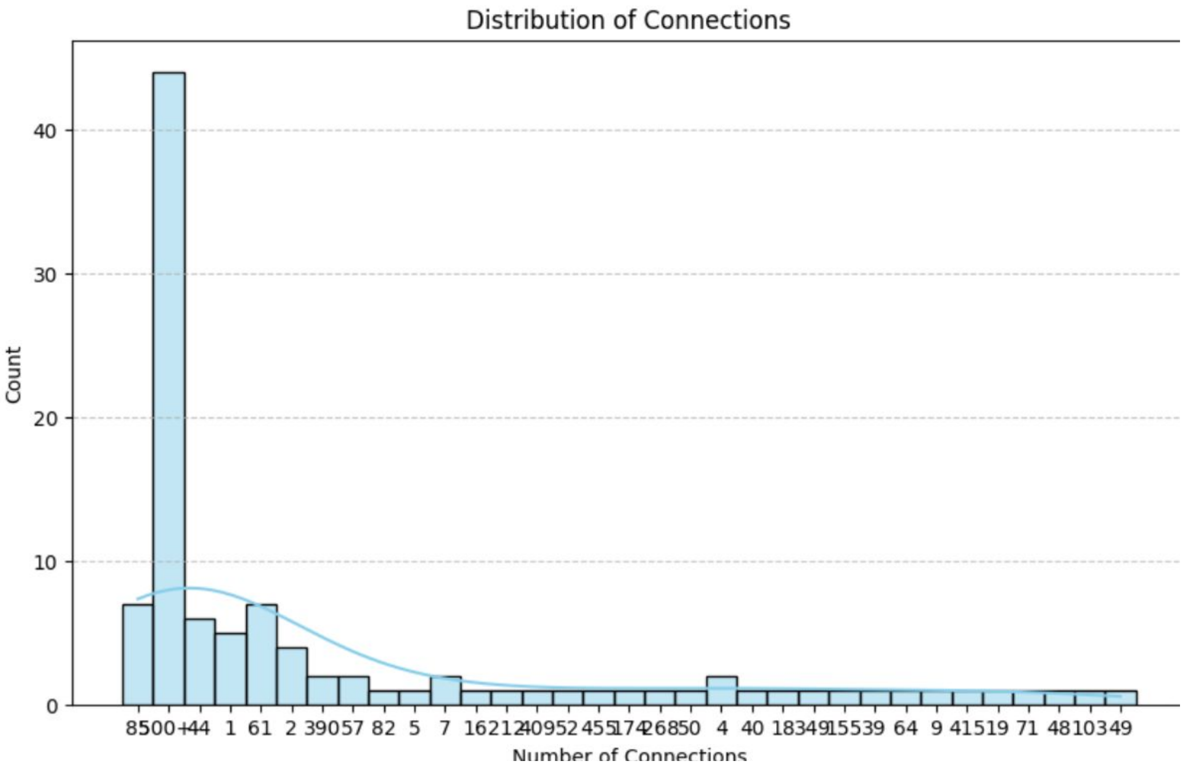
- **Dominant Human Resource**

EDA: Most Frequent Locations



- **Dominance of Specific Regions:** The 'Raleigh-Durham, North Carolina Area' stands out as the most frequent location, followed by 'Greater Philadelphia' and 'Houston, Texas'.

EDA: Connections



- **Dominant '500+' Category:** The most prominent bar is for '500+' connections. This indicates that a significant portion of the candidates in your dataset are highly networked,

EDA: Fit

```
df['fit']
```

	fit
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN
...	...
99	NaN
100	NaN
101	NaN
102	NaN
103	NaN

104 rows × 1 columns

dtype: float64

- No ground truth

Data preprocessing

1. ‘connection’ Column Transformation: From categorical to numeric for quantitative analysis and ranking tie-breaking
2. ‘job_title’ Text processing: standardise for effective text vectorization

Text Preprocessing

1. Convert to Lowercase
2. Tokenisation
3. Regex for Specific Patterns: Remove non-informative patterns eg. symbols, URLs, hashtags, etc
4. Stopword removal: Eliminate words like ‘a’. ‘the’, ‘is’
5. Punctuation removal: eg. ‘Engineer’ and ‘Engineer.’ are treated equally
6. Porter Stemming: eg. ‘aspiring’ -> ‘aspir’
7. Join Tokens to String: combine cleaned and stemmed words back into single string

Results

1. Bag of Words 'aspiring human resources'

	job_title	connections	similarity_score
76	Human Resources|\nConflict Management|\nPolici...	409	0.408248
73	Human Resources Professional	16	0.408248
72	Aspiring Human Resources Manager, seeking inte...	7	0.408248
27	Seeking Human Resources Opportunities	390	0.353553
29	Seeking Human Resources Opportunities	390	0.353553
88	Director Human Resources at EY	349	0.353553
96	Aspiring Human Resources Professional	71	0.353553
98	Seeking Human Resources Position	48	0.353553
2	Aspiring Human Resources Professional	44	0.353553
16	Aspiring Human Resources Professional	44	0.353553

- The Bag of Words matrix has a shape of (104, 173), meaning it processed 104 job titles and identified 173 unique features (words) after filtering.
- BoW effectively identifies lexical matches

2. TF IFD 'aspiring human resources'

	job_title	connections	similarity_score
76	Human Resources\nConflict Management\nPolici...	409	0.314119
73	Human Resources Professional	16	0.133722
72	Aspiring Human Resources Manager, seeking inte...	7	0.115721
96	Aspiring Human Resources Professional	71	0.113832
2	Aspiring Human Resources Professional	44	0.113832
16	Aspiring Human Resources Professional	44	0.113832
20	Aspiring Human Resources Professional	44	0.113832
32	Aspiring Human Resources Professional	44	0.113832
45	Aspiring Human Resources Professional	44	0.113832
57	Aspiring Human Resources Professional	44	0.113832

- **Top candidates are similar to the Bag of Words results.**
- **However, the similarity scores differ.**
- **TF-IDF gives more weight to words that are unique to a document but less frequent across the entire dataset**

3. Word2Vec 'aspiring human resources'

	job_title	connections	similarity_score
5	Aspiring Human Resources Specialist	1	0.876405
23	Aspiring Human Resources Specialist	1	0.876405
35	Aspiring Human Resources Specialist	1	0.876405
48	Aspiring Human Resources Specialist	1	0.876405
59	Aspiring Human Resources Specialist	1	0.876405
96	Aspiring Human Resources Professional	71	0.861600
2	Aspiring Human Resources Professional	44	0.861600
16	Aspiring Human Resources Professional	44	0.861600
20	Aspiring Human Resources Professional	44	0.861600
32	Aspiring Human Resources Professional	44	0.861600

- **Word2Vec should ideally capture semantic similarities better.**
- **The similarity scores might differ significantly from BoW/TF-IDF as the underlying vector representation is fundamentally different.**

4. Glove 'aspiring human resources'

	job_title	connections	similarity_score
5	Aspiring Human Resources Specialist	1	0.895304
23	Aspiring Human Resources Specialist	1	0.895304
35	Aspiring Human Resources Specialist	1	0.895304
48	Aspiring Human Resources Specialist	1	0.895304
59	Aspiring Human Resources Specialist	1	0.895304
96	Aspiring Human Resources Professional	71	0.892525
2	Aspiring Human Resources Professional	44	0.892525
16	Aspiring Human Resources Professional	44	0.892525
20	Aspiring Human Resources Professional	44	0.892525
32	Aspiring Human Resources Professional	44	0.892525

- GloVe, similar to Word2Vec, captures semantic relationships.
- eg. 'Specialist' and 'Professional' roles within Human Resources are grouped together
- HR Senior Specialist' and 'Advisory Board Member' roles, have very low or even negative similarity scores.

5. Fast Text 'aspiring human resources'

	job_title	connections	similarity_score
5	Aspiring Human Resources Specialist	1	0.831911
23	Aspiring Human Resources Specialist	1	0.831911
35	Aspiring Human Resources Specialist	1	0.831911
48	Aspiring Human Resources Specialist	1	0.831911
59	Aspiring Human Resources Specialist	1	0.831911
67	Human Resources Specialist at Luxottica	0	0.831911
76	Human Resources|\nConflict Management|\nPolici...	409	0.823783
72	Aspiring Human Resources Manager, seeking inte...	7	0.814095
87	Human Resources Management Major	18	0.806596
96	Aspiring Human Resources Professional	71	0.788740

- **While FastText excels at capturing semantic relationships, especially with its ability to handle out-of-vocabulary (OOV) words through subword information, its performance can depend on the pre-trained model and the specific dataset.**

6. BERT 'aspiring human resources'

	job_title	connections	similarity_score
5	Aspiring Human Resources Specialist	1	0.905480
23	Aspiring Human Resources Specialist	1	0.905480
35	Aspiring Human Resources Specialist	1	0.905480
48	Aspiring Human Resources Specialist	1	0.905480
59	Aspiring Human Resources Specialist	1	0.905480
96	Aspiring Human Resources Professional	71	0.902632
2	Aspiring Human Resources Professional	44	0.902632
16	Aspiring Human Resources Professional	44	0.902632
20	Aspiring Human Resources Professional	44	0.902632
32	Aspiring Human Resources Professional	44	0.902632

- As expected, BERT has done an excellent job of identifying highly relevant candidates. The top results are dominated by job titles like "Aspiring Human Resources Professional" and "Aspiring Human Resources Specialist," all with very high similarity scores (around 0.904-0.902).

7. SBERT 'aspiring human resources'

	job_title	connections	similarity_score
96	Aspiring Human Resources Professional	71	0.949807
2	Aspiring Human Resources Professional	44	0.949807
16	Aspiring Human Resources Professional	44	0.949807
20	Aspiring Human Resources Professional	44	0.949807
32	Aspiring Human Resources Professional	44	0.949807
45	Aspiring Human Resources Professional	44	0.949807
57	Aspiring Human Resources Professional	44	0.949807
5	Aspiring Human Resources Specialist	1	0.928035
23	Aspiring Human Resources Specialist	1	0.928035
35	Aspiring Human Resources Specialist	1	0.928035

- **Both generic BERT and SBERT still yield highly relevant results for this query.**

BERT on 'full stack software engineer'

	job_title	connections	similarity_score
65	Experienced Retail Manager and aspiring Human ...	57	0.748332
83	Human Resources professional for the world lea...	50	0.740886
85	Information Systems Specialist and Programmer ...	4	0.732927
70	Human Resources Generalist at ScottMadden, Inc.	0	0.711923
71	Business Management Major and Aspiring Human R...	5	0.701863
96	Aspiring Human Resources Professional	71	0.700898
2	Aspiring Human Resources Professional	44	0.700898
16	Aspiring Human Resources Professional	44	0.700898
20	Aspiring Human Resources Professional	44	0.700898
32	Aspiring Human Resources Professional	44	0.700898

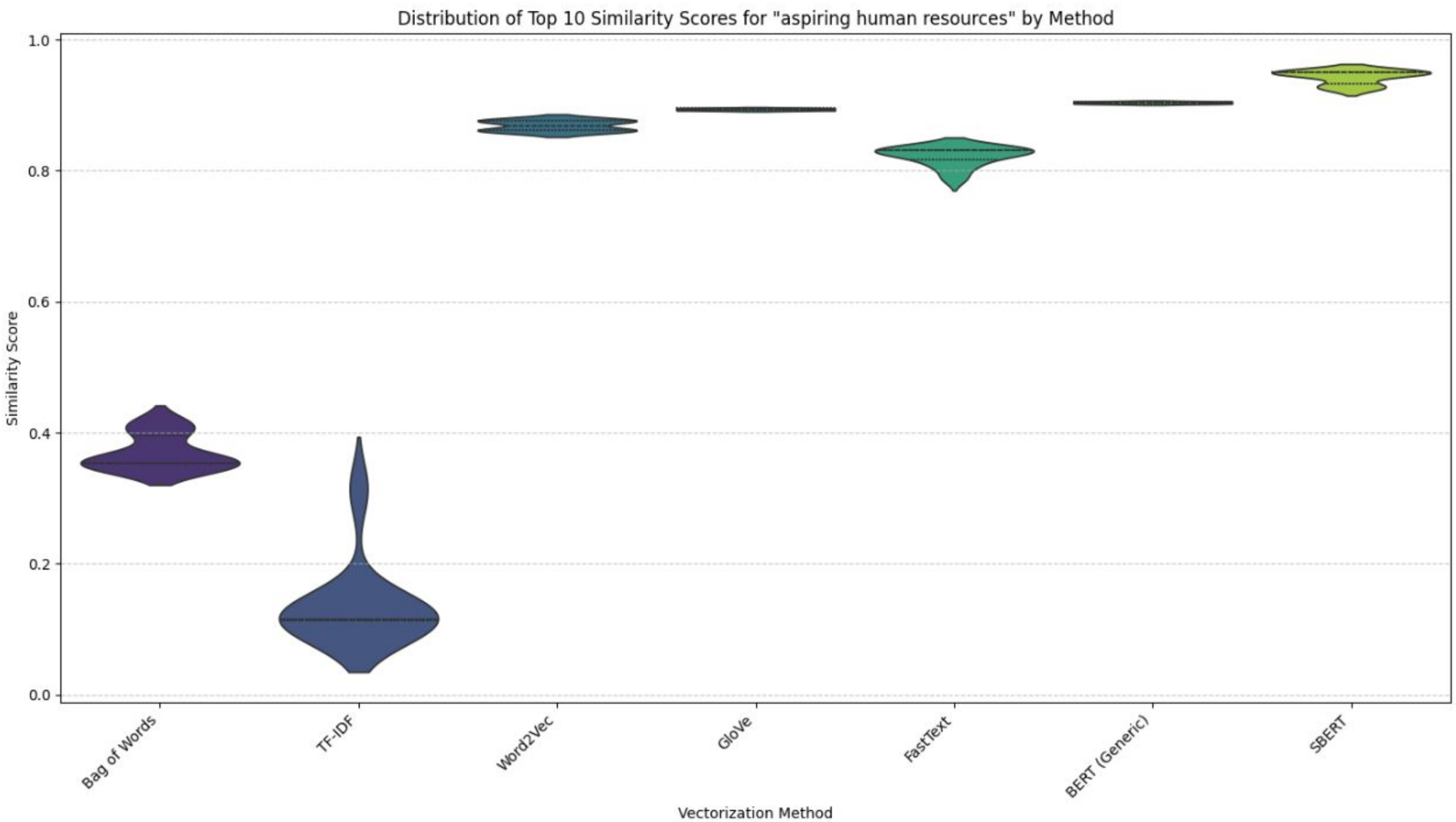
- **Both generic BERT and SBERT still yield highly relevant results for this query.**

SBERT II ‘full stack software engineer’

	job_title	connections	similarity_score
85	Information Systems Specialist and Programmer ...	4	0.500883
79	Junior MES EngineerI Information Systems	52	0.446146
83	Human Resources professional for the world lea...	50	0.428814
73	Human Resources Professional	16	0.324346
3	People Development Coordinator at Ryan	0	0.316935
17	People Development Coordinator at Ryan	0	0.316935
21	People Development Coordinator at Ryan	0	0.316935
33	People Development Coordinator at Ryan	0	0.316935
46	People Development Coordinator at Ryan	0	0.316935
58	People Development Coordinator at Ryan	0	0.316935

- **SBERT's fine-tuning** for semantic similarity yields more reliable and relevant rankings by providing more semantically meaningful sentence embeddings, leading to better distinction between categories.

Model Comparison



- **SBERT and BERT (Generic)** consistently show the highest and most tightly clustered similarity scores
- **Word2Vec, Glove, FastTex:** Lower similarity scores than SBERT and BERT
- **TF-IDF and Bag of Words (BoW)** generally exhibit lowest similarity scores and potentially wider distributions, particularly for TF-IDF.

Model comparison

Method	Meaning	Context	Level	Best Use
BoW	❌	❌	Document	Baseline
TF-IDF	⚠️	❌	Document	Search
Word2Vec	✅	❌	Word	Semantics
GloVe	✅	❌	Word	Pretrained
FastText	✅	❌	Word	Noisy text
BERT	✅	✅	Token	Deep NLP
SBERT	✅	✅	Sentence	Similarity

Demo

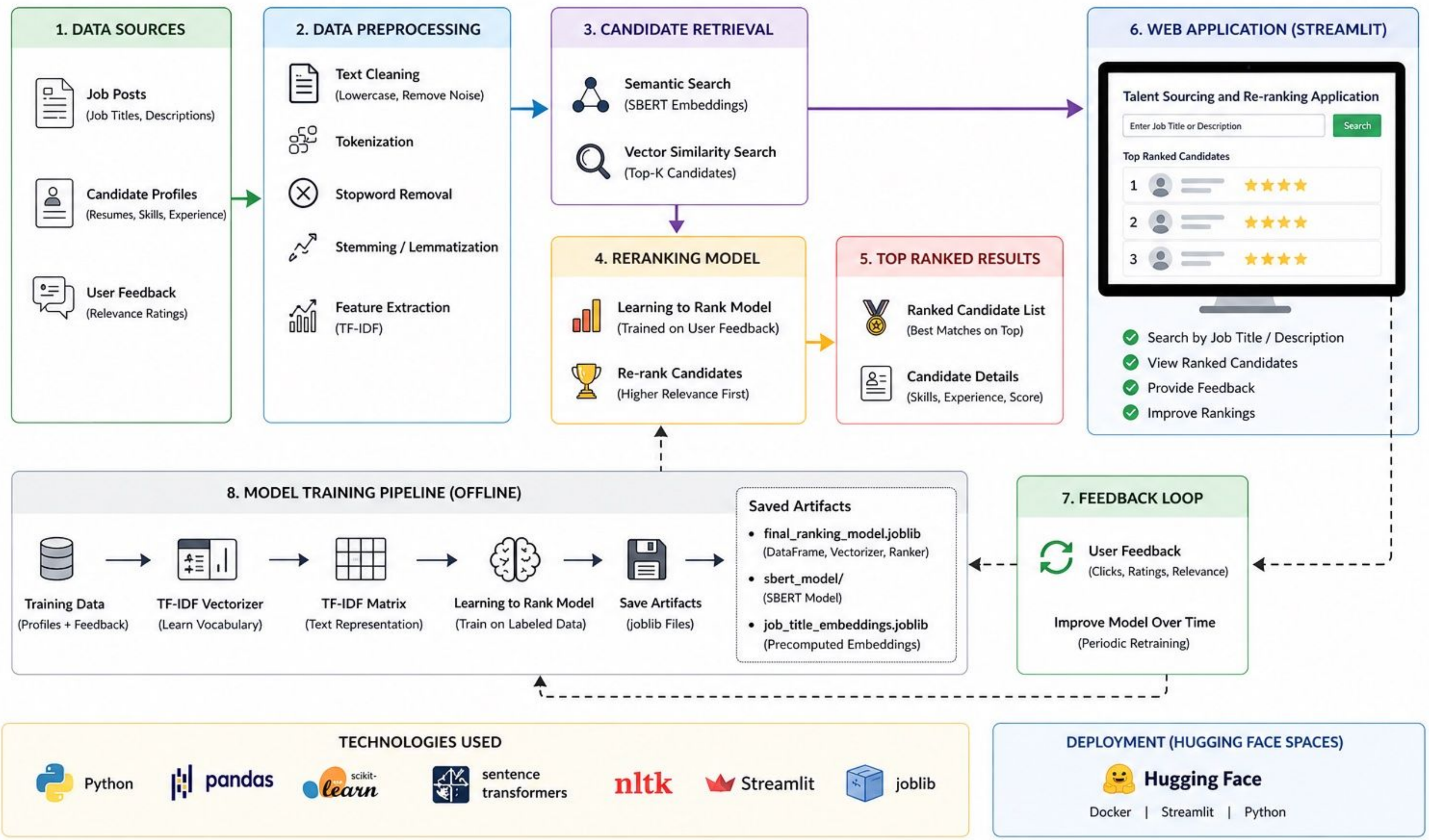
Tools used

- Python
- Jupyter/Google Colab
- Pandas
- NumPy
- Scikit-learn: machine learning utilities, including TfidfVectorizer and cosine_similarity.
- NLTK: For NLP tasks like tokenization and stemming.
- Gensim: For Word2Vec and FastText model handling.
- BERT and SBERT embeddings.
- Matplotlib & Seaborn: For data visualization.
- Streamlit: For building the interactive web application.
- Hugging Face Spaces: For deploying the web application

Future improvements

- **Advanced Semantic Models:** LLMs, finetuning existing ones
- **Multi-Modal Data Integration:** Incorporate other data types beyond job titles
- **Personalization:** Allow individual recruiters to fine-tune the ranking model based on their specific preferences or success metrics for different roles.
- **Dynamic Thresholding:** cut-off points that dynamically adjust based on the query's complexity, the distribution of similarity scores, or historical success rates for similar roles.
- **Explainability Features in UI:** Enhance the Streamlit application to visually explain why a candidate was ranked in a certain position, showing contributing terms or semantic relationships.
- **Scalability:** Optimize the solution for larger datasets and higher query volumes, potentially by using specialized vector databases or efficient indexing techniques.
- **Evaluation Metrics:** Establish robust offline evaluation metrics (e.g., precision, recall, NDCG) to systematically compare different models and ranking strategies, especially as more feedback data becomes available

Summary



1. DATA SOURCES
Job Posts
(Job Titles, Descriptions)
Candidate Profiles
(Resumes, Skills, Experience)
User Feedback
(Relevance Ratings)
2. DATA PREPROCESSING
Text Cleaning
(Lowercase, Remove Noise)
Tokenization
Stopword Removal
Stemming / Lemmatization
Feature Extraction
(TF-IDF)
3. CANDIDATE RETRIEVAL
Semantic Search
(SBERT Embeddings)
Vector Similarity Search
(Top-K Candidates)
4. RERANKING MODEL
Learning to Rank Model
(Trained on User Feedback)
Re-rank Candidates
(Higher Relevance First)
5. TOP RANKED RESULTS
Ranked Candidate List
(Best Matches on Top)
Candidate Details
(Skills, Experience, Score)
6. WEB APPLICATION (STREAMLIT)
Talent Sourcing and Re-ranking Application
Enter Job Title or Description
Search
Top Ranked Candidates
1
2
3
Search by Job Title / Description
View Ranked Candidates
Provide Feedback
Improve Rankings
8. MODEL TRAINING PIPELINE (OFFLINE)
Training Data
(Profiles + Feedback)
TF-IDF Vectorizer
(Learn Vocabulary)
TF-IDF Matrix
(Text Representation)
Learning to Rank Model
(Train on Labeled Data)
Save Artifacts
(joblib Files)
Saved Artifacts
final_ranking_model.joblib
(DataFrame, Vectorizer, Ranker)
sbert_model/
(SBERT Model)
job_title_embeddings.joblib
(Precomputed Embeddings)
7. FEEDBACK LOOP
User Feedback
(Clicks, Ratings, Relevance)
Improve Model Over Time
(Periodic Retraining)
TECHNOLOGIES USED
Python
pandas
scikit-learn
sentence transformers
nltk
Streamlit
joblib
DEPLOYMENT (HUGGING FACE SPACES)
Hugging Face
Docker | Streamlit | Python

Thank You

- Thank you for your time
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